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Effects of Montmorillonite on Crude Oil Biodegradation and the Microbial Community in an Oil Production Well Pad Shut Down for 753 Days

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Abstract

Clay-mediated microbial degradation has been demonstrated as a low-cost, efficient, and eco-friendly strategy for remediating crude oil-contaminated soils. Despite substantial laboratory studies, field tests remain scarce. In this study, montmorillonite treatment was applied to a crude oil production well pad shut down for 753 days. Post-treatment analyses included soil physicochemical parameters (water content, redox potential, pH, elemental analysis, and total organic carbon), crude oil content/composition (gas chromatography–mass spectrometry), microbial biomass (deoxyribonucleic acid concentration), and community structure (high-throughput sequencing), with parallel comparisons to untreated control areas. Results indicated that montmorillonite enhanced the crude oil biodegradation rate (37.27% vs. control 33.00%), increased microbial biomass (83.08% vs. control 35.06%), and enriched biodiversity (7 genera vs. control 0). Specifically, it exhibited the most pronounced promotion effects on saturated hydrocarbon degradation (73.42% vs. control 60.89%) and aromatic hydrocarbon degradation (45.77% vs. control 29.60%). This study not only provides field evidence for clay-mediated microbial remediation but also lays a foundation for developing composite remediation approaches (e.g., nutrient supplements, catalysts, or functional microbial consortia) in future research and practical applications.

Keywords: montmorillonite; crude oil; microbial degradation; microbiome succession; saturated hydrocarbons; aromatic hydrocarbons

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Academic Editors: Gianniantonio Petruzzelli and Gabriel Gascó Guerrero

Received: 20 October 2025
Revised: 24 December 2025
Accepted: 29 December 2025
Published: 31 December 2025

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1. Introduction

Crude oil, as an indispensable strategic resource for modern societal development, plays a pivotal role in global industrial operations, economic advancement, and innovation-driven growth. However, crude oil pollution poses severe environmental challenges, persistently threatening ecosystems and public health [1]. Industrial and petroleum activities are estimated to accidentally release 1.7×10^6–8×10^6 metric tons of oil into the environment annually [2], leaving numerous contaminated sites requiring remediation. Their accumulation in soil disrupts soil structure and fertility, leading to land degradation, reduced crop yields, and biodiversity loss [3]. Recent studies have confirmed significant contaminant migration between groundwater and surrounding soils [4,5].

Among various crude oil-contaminated soil remediation methods, clay-mediated microbial degradation of crude oil has emerged as one of the most promising approaches due to its ecological friendliness, cost-effectiveness, and demonstrated efficacy. Clay

minerals typically possess significant characteristics such as large specific surface area [6], high cation exchange capacity [7], swelling behavior [8,9], and unique microstructure [10], which collectively create favorable conditions for microbial degradation. These appropriate properties enable clay minerals to provide nutrients, physical protection, microenvironment regulation (moisture and pH control), and metabolic enhancement (catalysis and electron transfer) [11]. Furthermore, clay surfaces serve as attachment sites for hydrocarbons, further facilitating their microbial degradation.

Current research has investigated the effects of various clay minerals including montmorillonite [12–21], kaolinite [12,15,16,21–24], illite [25], nontronite [26], vermiculite [27], palygorskite [15,16,19,21,28,29], and saponite [15,16,20,21,30] on crude oil microbial degradation. Among these, montmorillonite has received extensive attention due to its maximum specific surface area, high cation exchange capacity, and swelling properties. However, these reports have been limited to laboratory studies, with no field tests conducted using solely montmorillonite for microbial remediation of crude oil-contaminated soil at crude oil production well pads [31]. Actual well sites exist in open environments where microbial degradation of crude oil is influenced by complex climatic conditions (temperature, wind, sunlight, and precipitation). The field test results of montmorillonite-mediated microbial remediation would provide crucial reference information for field applications and future research.

This study presents the first field test of montmorillonite-mediated microbial remediation of crude oil-contaminated soil at a long-term shutdown well pad of production. The field test compared physicochemical parameters, crude oil content and composition, microbial biomass, and community structure between montmorillonite-treated and natural attenuation zones. The field test lasted 753 days, with no additional interventions beyond montmorillonite application and plowing.

2. Materials and Methods

2.1. Field Test Well Pad

The field test well pad of this study is located in the Ordos Basin, operated by Shaanxi Yangchang Petroleum (Group) Co., Ltd. (Yan'an, China). The ground elevation exceeds 1300 m (As per the confidentiality agreement, the actual name and precise data cannot be provided). The surface is characterized by crisscrossed gullies and belongs to the loess tableland geomorphology, covered by 100–200-meter-thick Quaternary loess deposits. The region experiences a temperate semi-arid monsoon climate, with an annual average temperature of 7–10 °C. The average temperature in January ranges from −5 °C to −9 °C, while July temperatures typically reach 20–23 °C. The extreme minimum temperature recorded is −28 °C. Annual precipitation averages 400–500 mm, predominantly concentrated between July and September. Summers frequently encounter heavy rainstorms, whereas winters and springs are arid, accompanied by sandstorms and cold waves. According to operational requirements, surface vegetation was cleared, and ground hardening (compaction) was implemented within the well pad area.

This well pad contains 8 production wells (Figure S1). Their primary oil-producing layer is the Triassic Yanchang Formation, with an average burial depth of approximately 1900 m, an effective thickness of about 30 m, a reservoir temperature of around 65 °C, and a reservoir pressure of around 15 MPa. The target formation is an ultra-low permeability reservoir, with crude oil density of 0.85 g/cm^3 (28 °C) and viscosity of 2.66 mPa·s (20 °C). These 8 wells were put into production in 2002. In May 2021, they were shut down as part of a macro-adjustment strategy to restore reservoir pressure. They resumed production in June 2023. During the shutdown period, only routine inspections were conducted, with no operations or leaks that could have increased soil crude oil content at the well pad.

2.2. Field Test Methods

On the day of well pad shutdown, following the completion of shutdown operations (including retrieval of critical equipment and locking of mechanical devices), initial soil samples were collected from a 15 cm wide circular zone around the wellheads of the 8 wells (Figure S1). The rationale for this procedure is that spilled crude oil primarily originates from wellhead operations, such as packing replacement and rod string maintenance. Additionally, the presence of valves and sampling ports at wellheads increases the likelihood of leakage. Consequently, this area is typically identified as the most heavily contaminated within the entire well pad.

Following the collection of initial samples, montmorillonite treatment was implemented. On one side, 60.2 tons of montmorillonite were uniformly distributed over a 2304 m^2 area surrounding 4 wells, followed by plowing to a depth of approximately 15 cm to ensure thorough mixing with the topsoil and subsequent land leveling, constituting the test group (Figure S1). On the opposite side, no montmorillonite was applied to the area around the other 4 wells, which underwent plowing (15 cm depth) and leveling as the control group (Figure S1).

After 753 days, all 8 wells were resampled using the identical methodology. Soil samples from the 4 treated wells were homogenized to form the test composite sample (before and after field test were labeled as M0 and M753, respectively), while those from the 4 untreated wells were similarly homogenized into the control composite sample (before and after field test were labeled as C0 and C753, respectively).

2.3. Properties of Pristine Soil and Montmorillonite

The natural pristine soils at the well pad and surrounding areas consist of Quaternary loess, predominantly chestnut soil (quartz and feldspar content > 60%, $CaCO_3$ 10–30%, organic matter 2–4%; illite and montmorillonite < 0.4% with minimal soluble salts) and aeolian sandy soil (quartz-dominated with minor calcium carbonate and organic matter). These soils exhibit low nutrient availability and poor pedogenic development. The peripheral zones outside the well pad (where vegetation is prohibited within the operational area according to safety protocols) feature sparse vegetation, mainly zonal steppe species including *Stipa bungeana*, *Thymus serpyllum*, and *Stipa breviflora*. The shrub layer predominantly comprises artificially planted *Hippophae rhamnoides Linn*, *Salix cheilophila*, and *Caragana sinica (Buchoz) Rehd*.

The montmorillonite was procured from Xinyang Licheng New Materials Technology Co., Ltd. (Xinyang, China), with the following chemical composition (wt%): Na_2O (0.05), MgO (6.70), Al_2O_3 (19.92), SiO_2 (58.33), P_2O_5 (0.02), K_2O (0.19), CaO (2.95), MnO (0.17), TiO_2 (0.25), Fe_2O_3 (1.77), and loss-on-ignition (9.17). The specific surface area of this montmorillonite is 688.79 m^2/g, and its cation exchange capacity is 116.43 cmol(+)/kg. The data on the chemical composition, specific surface area, and cation exchange capacity of the aforementioned montmorillonite were provided by Xinyang Licheng New Materials Technology Co., Ltd.

2.4. Measurement of Soil Physicochemical Parameters

2.4.1. Water Content

A 40 g soil sample was dried in an oven at 105 ± 5 °C until constant weight was achieved to determine the soil water content [32].

2.4.2. Conductivity

A 20 g soil sample was added to 100 mL deionized water, shaken for 30 min at 20 °C, and filtered to obtain a soil extract [33]. The extract's conductivity was measured using

an InLab 731 probe connected to a Mettler Toledo SevenMulti™ meter (Mettler Toledo, Columbus, OH, USA).

2.4.3. Redox Potential

The redox potential was measured directly on the soil sample using a platinum electrode and a saturated calomel electrode via the potential difference method [34]. Measurements were performed with an InLab Redox probe mounted on a Mettler Toledo SevenMulti™ meter (Mettler Toledo, Columbus, OH, USA).

2.4.4. pH

A 20 g soil sample was suspended in 50 mL deionized water, shaken for 30 min at 20 °C [35], and the pH of the suspension was measured using an InLab Expert Pro pH electrode coupled to the Mettler Toledo SevenMulti™ meter (Mettler Toledo, Columbus, OH, USA).

2.4.5. Total Organic Carbon (TOC)

A 0.05 g air-dried soil sample (sieved through a 160-mesh, 0.097 mm sieve) was treated with phosphoric acid solution (59 mL of 85% concentrated H_3PO_4 diluted in 941 mL deionized water) until gas evolution ceased [36]. The sample was then combusted in a 5000TOCi analyzer (Mettler Toledo, Columbus, OH, USA).

2.4.6. Elemental Analysis of C, H, O, and N

A 5 mg soil sample, homogenized by mortar grinding and dried at 60 °C for 4 h [37], was analyzed using a FlashSmart elemental analyzer (Thermo Fisher Scientific, Waltham, MA, USA).

2.5. Analysis of Crude Oil in Soil

2.5.1. Extraction of Crude Oil from Soil

The soil sample was homogenized with a 6-fold volume of organic solvent mixture (n-hexane (C_6H_{14}) and acetone (C_3H_6O) at 3:1 v/v) [38] and subjected to ultrasonication (60 min, 40 °C) for crude oil extraction. Subsequently, Soxhlet extraction was performed using dichloromethane (CH_2Cl_2) as the solvent at a water bath temperature of 78 °C [39] until the CH_2Cl_2 leachate exhibited no absorption in the 200–400 nm spectral range, as determined by a UV-Vis spectrophotometer (Varian Cary 100, Agilent, Santa Clara, California, USA) [40]. The crude oil dissolved in the organic solvent mixture and CH_2Cl_2 was collected, concentrated to 4 mL via rotary evaporation (Yarong, Shanghai, China) at 65 °C, and further dried to a constant weight at room temperature (24 °C) [40]. The weight was recorded every 4 h, with differences between three consecutive measurements maintained within 0.0010 g to ensure precision.

2.5.2. Fractions Analysis of Crude Oil

The fractions of saturated hydrocarbons, aromatic hydrocarbons, resins, and asphaltenes (SARA) in crude oil were separated via column chromatography based on differences in polarizability and polarity. Briefly, 20.0–50.0 mg of heavy oil was dissolved in 30 mL of n-hexane, followed by ultrasonication (5 min) and filtration through absorbent cotton to isolate the asphaltenes. The remaining fractions were sequentially eluted using a chromatographic column packed with 4 g of silica gel and 3 g of activated alumina, with n-hexane, dichloromethane, and ethanol as mobile phases [41].

2.5.3. Gas Chromatography–Mass Spectrometry (GC-MS) Analysis of Saturated and Aromatic Hydrocarbons

The compositional analysis of saturated hydrocarbons and aromatic hydrocarbons in crude oil was conducted via GC-MS [42]. Deuterated tetracosane (D_{50}-nC_{24}, 10 μg) and deuterated dibenzothiophene (D_8-dibenzothiophene, 10 μg) served as internal standards for saturated hydrocarbons and aromatic hydrocarbons quantification, respectively [43].

The GC-MS system comprised a Trace-DSQ mass spectrometer (Thermo Finnigan, San Jose, CA, USA) interfaced with an HP 6890 gas chromatograph (Agilent, Santa Clara, CA, USA). Separation was achieved using an HP-5MS capillary column with helium (99.99% purity) as the carrier gas. The temperature program initiated at 50 °C, then ramped to 120 °C (20 °C/min), 250 °C (4 °C/min), and finally 310 °C (3 °C/min), holding for 30 min. The mass spectrometer operated in full-scan electron ionization mode at 70 eV [44].

2.6. Analysis of Microorganisms in Soil

2.6.1. Extraction of Soil DNA

Soil samples (0.5 g) were subjected to mechanical lysis through vortex mixing (2200 rpm, 5 min) in the presence of 600 μL extraction buffer (50 mM sodium phosphate buffer, 50 mM NaCl, 500 mM Tris-HCl, and 5% sodium dodecyl sulfate), 300 μL phenol-chloroform-isoamylic alcohol (25:24:1, v/v), and 0.5 g sterile 0.5 mm glass beads. The resulting lysate was centrifuged at 16,000 g for 2 min. The aqueous supernatant was subsequently re-extracted with an equal volume of phenol-chloroform-isoamylic alcohol (25:24:1, v/v) followed by centrifugation at 16,000 g for 5 min. The upper aqueous phase was collected and subjected to further purification through mixing with an equivalent volume of chloroform-isoamylic alcohol (24:1, v/v) and centrifugation at 16,000 g for 5 min. Nucleic acids were precipitated from the final supernatant by adding 1 volume of ice-cold isopropanol and incubating at −20 °C for 20 min [45]. DNA quantification was performed using a Qubit fluorometer (Invitrogen, Thermo Fisher Scientific, Waltham, MA, USA) with reference to standard calibration curves [46].

2.6.2. Gene Sequence Analysis of Soil

DNA from the soil samples was analyzed by the 16 s rRNA gene sequence using the forward primer GM3F (5′-AGAGTTTGATCMTGGC-3′) and reverse primer GM4R (5′-TACCTTGTTACGACTT-3′) [45]. The genes were sequenced on a MiSeqTM System (Illumina, San Diego, DA, USA) [47] and analyzed using the Galaxy Platform [48] (https://galaxyproject.org/). We compared the sequencing results with Gene-Bank or Eztaxon.

2.7. Analysis of Microorganisms in Crude Oil

2.7.1. DNA Extraction of Crude Oil

The crude oil was subjected to extraction with 2,2,4-trimethylpentane (isooctane) and homogenized with 3–4 mm glass beads using vortex mixing. The mixture was then centrifuged at 7870 g for 20 min at 4 °C. Following centrifugation, the supernatant was carefully removed while retaining the pelleted fraction [49]. Genomic DNA was subsequently quantified using a QIAamp DNA Stool Mini Kit (Hilden, North Rhine-Westphalia, Germany) [50] with fluorescence detection performed on an Invitrogen Qubit 4.0 Fluorometer (Thermo Fisher Scientific, Carlsbad, California, USA) [46].

2.7.2. Gene Sequence Analysis of Crude Oil

DNA from the crude oil was analyzed by the 16 s rRNA gene sequence using the forward primer 341 f (5′-CCTACGGAGGCAGCAGCAG-3′) and reverse primer 518 r (5′-ATTACCGCGCTGCTGG-3′) [51]. The genes were sequenced on a MiSeqTM System

(Illumina, San Diego, DA, USA) [47], analyzed using the Galaxy Platform [48] (https://galaxyproject.org/), and compared the sequencing results with Gene-Bank and Eztaxon.

All chemical agents used in this study, unless otherwise specified, were of an analytical grade and supplied by Tianjin Fuchen Chemical Reagents Factory, Tianjin, China.

3. Results

3.1. Physicochemical Properties of Well Pad Soil

3.1.1. Soil Water Content

Water content of well pad soil is controlled by precipitation. The well pad belongs to a temperate semi-arid monsoon climate. Although the precipitation is not high, the evaporation is also not high. According to the four samples, the soil water content (8.44–12.97%) supports microbial degradation activities (Table 1). The higher water content observed after 753 days may be associated with precipitation at the time of sampling.

Table 1. Physicochemical properties of soil in the test and control groups before and after the field test.

Sample	Water Content (%)	Conductivity (μs/cm)	Redox Potential (mV)	pH	TOC (%)	Crude Oil Content (mg/kg)	DNA Concentration (μg/g)
M0	8.44 ± 0.14	301 ± 17	241.5 ± 1.2	7.99 ± 0.01	7.44 ± 0.14	69,146 ± 84	65 ± 4
M753	12.11 ± 0.21	213 ± 9	213.0 ± 0.2	7.44 ± 0.01	4.93 ± 0.19	43,372 ± 92	119 ± 2
C0	8.91 ± 0.16	290 ± 9	242.7 ± 0.6	7.96 ± 0.01	6.81 ± 0.10	66,981 ± 73	77 ± 3
C753	12.97 ± 0.09	239 ± 12	226.9 ± 0.6	7.18 ± 0.01	4.85 ± 0.07	44,875 ± 106	104 ± 5

Note: Measure three replicate samples to determine the error.

3.1.2. Soil Conductivity

The observed soil conductivity range (200–300 μS/cm) reflects both low salinity and inherently poor fertility. Notably, this salinity level imposes no inhibitory effects on crude oil biodegradation by microorganisms (Table 1).

3.1.3. Soil Redox Potential

The redox potential values indicate that the well pad soil is under typical weakly oxidizing to weakly reducing conditions (Table 1). Moreover, the lower redox potential observed at the end of the field test corresponds with the higher water content.

3.1.4. Soil pH

Despite the well pad soil maintaining an alkaline pH range in general, a consistent acidification tendency was detected in both montmorillonite-treated and untreated zones. Notably, montmorillonite application demonstrated a buffering effect against soil acidification (Table 1).

3.1.5. Soil TOC

The soil surrounding the oil production wellhead exhibited high TOC content. After 753 days, a declining trend was observed in both montmorillonite-treated and untreated zones, with reductions of 33.74% and 28.78%, respectively (Table 1).

3.1.6. Soil C, H, O, and N

The contents of N and O elements in the soil showed no obvious changes before and after the field test. The reductions in C and H in the test group (M753) were 25.6% and 52.2%, respectively, while those in the control group (C753) were 24.5% and 41.7%, respectively (Figure 1). This indicates stronger crude oil degradation in the test group. The decline in H/C ratio suggests preferential microbial degradation of saturated hydrocarbon

components. Incomplete biodegradation or accumulation of oxygen-containing compounds (e.g., intermediate metabolites) resulted in minimal changes in the O/C ratio (Figure 1).

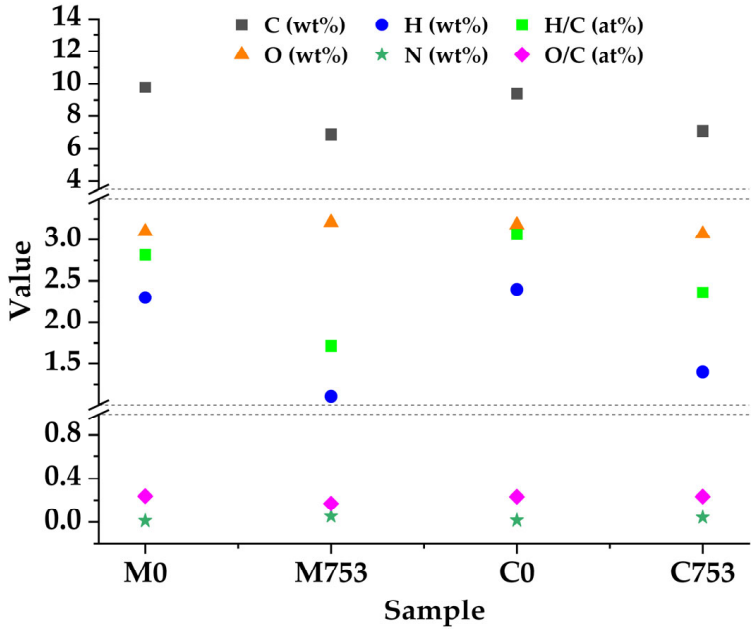

Figure 1. Content of C, H, O, and N before and after field test.

3.2. Crude Oil in Soil

After the field test, the crude oil concentration in the test group decreased from 69,146 mg/kg to 43,372 mg/kg, while in the control group it declined from 66,981 mg/kg to 44,875 mg/kg (Table 1).

3.2.1. SARA Fractions

The SARA analysis indicated that the crude oil consisted of 71% saturated hydrocarbons, 18% aromatics hydrocarbons, 8% resins, and 2% asphaltenes (Figure S2). The SARA composition of crude oil in the soil samples at the beginning of the field test (M0 and C0) was similar to that of reservoir crude oil (Figure S2), likely due to fresh oil entering the soil during well shutdown operations (Figure S3). By the end of the field test, changes were observed in the SARA composition of the crude oil in the soil. In sample M753, the SARA composition is 46% saturated hydrocarbons, 27% aromatic hydrocarbons, 22% resins, and 5% asphaltenes. For C753, saturated hydrocarbons account for 53%, aromatic hydrocarbons 26%, resins 17%, and asphaltenes 4% (Figure S2). This variation results from the preferential biodegradation of saturated and aromatic hydrocarbons in crude oil by indigenous soil microorganisms. Given the extremely slow microbial degradation rate of asphaltenes, their content (2%) can be used as a baseline to calculate the natural degradation rates of other components (Figure 2). In the test group, the biodegradation rates of saturated and aromatic hydrocarbons are 52.2% and 9.2%, respectively, while in the control group, they reach 43.9% and 5.8% (Figure 2).

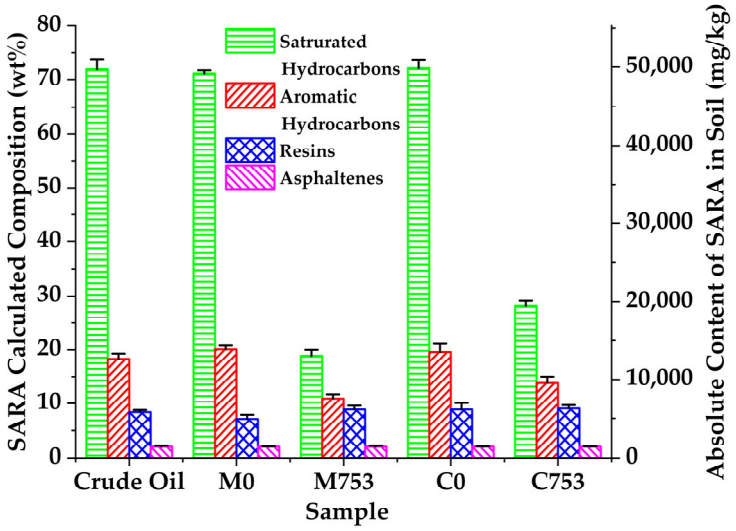

Figure 2. Calculated composition and absolute content of SARA in crude oil in well pad soil samples before and after field test.

3.2.2. Saturated Hydrocarbons

The fraction of saturated hydrocarbons was further analyzed by GC-MS, and a comparison with the crude oil revealed that the saturated hydrocarbon components in M0 and C0 remained relatively intact, confirming only minor biodegradation of the crude oil in the soil prior to the field test (Figure S4). Through peak integration of the GC-MS data from the test group and the control group, the concentration changes of 27 n-alkanes, 25 alkylcyclohexanes, 24 terpenoids, 16 hopanes, and 20 steranes in the saturated hydrocarbons were determined.

1. n-Alkanes

The n-alkanes with carbon numbers below C_{20} in both M0 and C0 samples exhibited noticeable depletion (Figure 3), which may result from rapid volatilization. Prior to the field test, the soil samples displayed complete n-alkane sequences without detectable missing homologues (Figure 3). Following the field test, obvious reductions were observed in n-alkanes below C_{31} for both M753 and C753 samples, with the montmorillonite-treated soil (M753) demonstrating more pronounced concentration decreases compared to the control (C753). Faint variations were detected in n-alkanes beyond C_{32} when comparing pre- and post-test conditions. Notably, the C_{12} and C_{13} n-alkanes were completely eliminated in M753 and C753 samples, attributable to the combined effects of volatilization and biodegradation processes (Figure 3).

2. Alkylcyclohexanes

A consistent decrease in total alkylcyclohexanes (C_6–C_{30}) was observed in both test and control groups following completion of the field test. Montmorillonite exhibited mild stimulation of microbial degradation activity (Figure 4).

3. Terpanes

For bicyclic sesquiterpenes, natural degradation was observed (C753), with montmorillonite exhibiting a stimulating effect on their degradation (M753). In contrast, tricyclic terpanes showed minimal natural degradation by soil microorganisms (C753), even in the presence of montmorillonite (M753) (Figure 5).

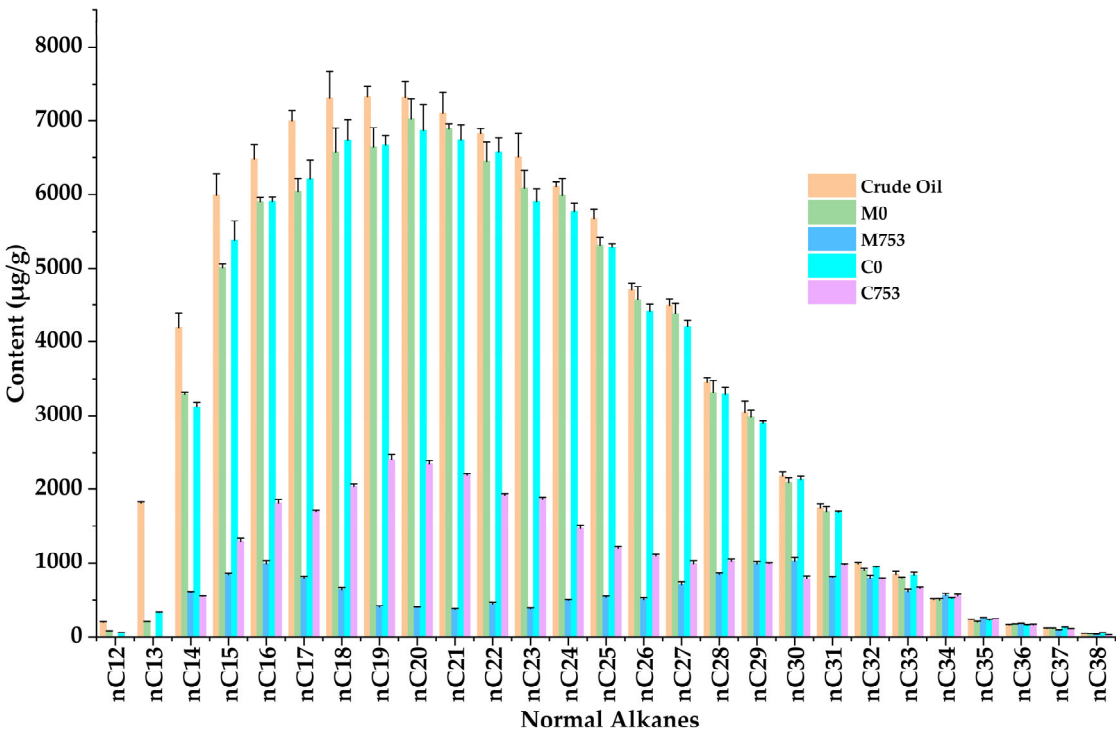

Figure 3. Content of n-alkanes in crude oil-contaminated soil from well pad before and after field test.

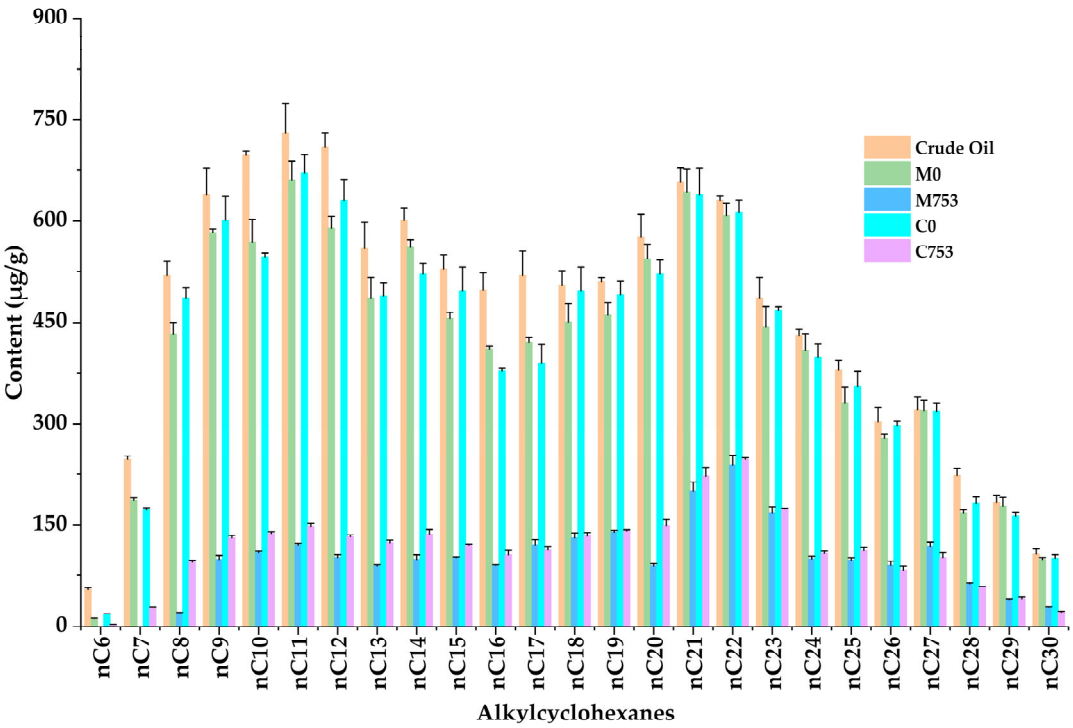

Figure 4. Content of alkylcyclohexanes in crude oil-contaminated soil from well pad before and after field test.

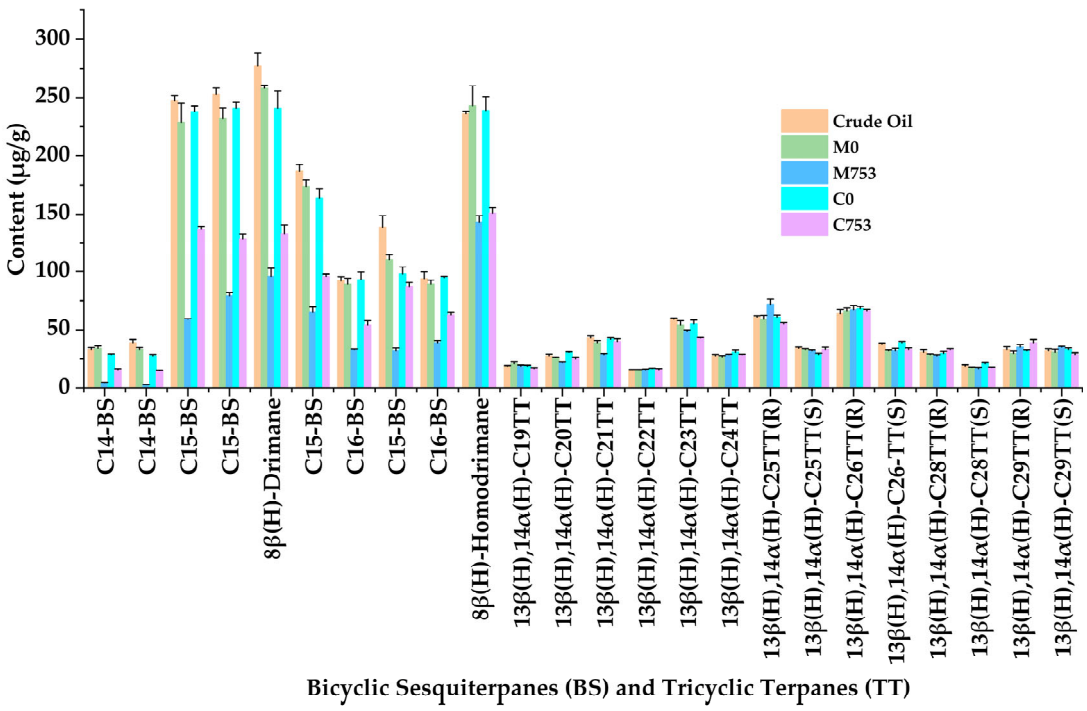

Figure 5. Content of bicyclic sesquiterpanes and tricyclic terpanes in crude oil-contaminated soil from well pad before and after field test.

4. Hopanes

The crude oil, M0, M753, C0, and C753 exhibited similar hopane concentrations, indicating that indigenous microorganisms in the well pad soil were incapable of effectively biodegrading hopanoid compounds through natural attenuation (C753) (Figure S5). The addition of montmorillonite failed to stimulate microbial biodegradation of hopane series (Figure S5).

5. Steranes

The indigenous microbial community is capable of naturally degrading sterane series compounds (Figure S6). The influence of montmorillonite on sterane degradation is weak, with both stimulation and inhibition effects being weak and not exhibiting stable or consistent patterns (Figure S6).

3.2.3. Aromatic Hydrocarbons

Overall, before the field test, the aromatic hydrocarbons C0 and M0 already showed missing components, exhibiting clear biodegradation characteristics compared to the crude oil (Figure S7), which differs from the case of saturated hydrocarbons (Figures S4 and S7). After the field test, further degradation could be observed (C753 and M753) (Figure S7). Below, the changes in aromatic hydrocarbon content are discussed separately for naphthalene, phenanthrene, fluorene, and biphenyl series compounds, as well as high-ring number (≥4) aromatic hydrocarbons.

6. Naphthalene series

The naphthalene series compounds exhibited a concentration pattern of crude oil > M0 ≈ C0 > C753 > M753, indicating that soil microorganisms can degrade naphthalene series compounds and that montmorillonite stimulates this biodegradation process (Figure 6).

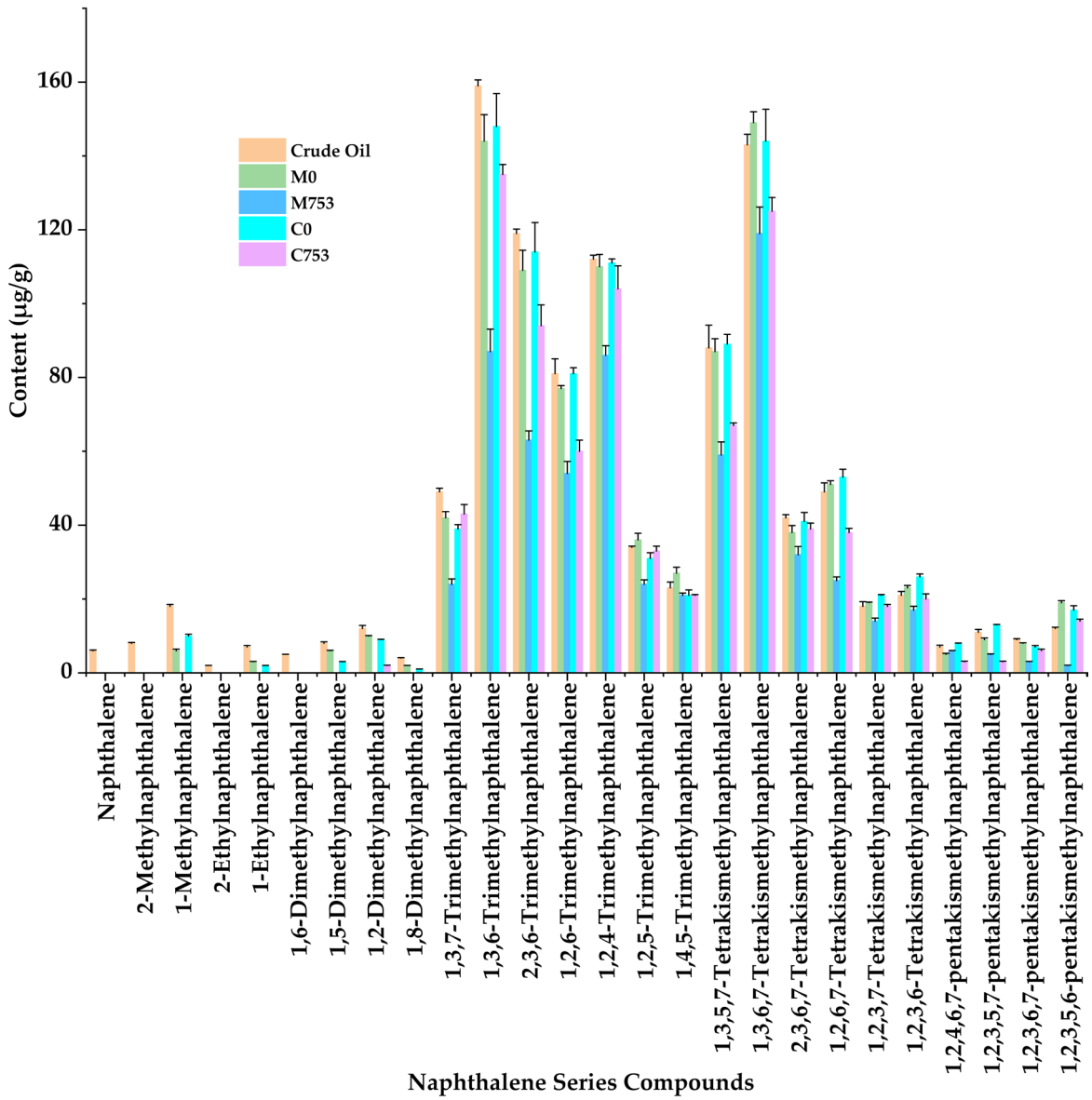

Figure 6. Content of naphthalene series compounds in crude oil-contaminated soil from well pad before and after field test.

7. Phenanthrene series

Natural biodegradation is limited to partial phenanthrene series compounds, such as phenanthrene, monomethylphenanthrene, and certain dimethylphenanthrenes, with montmorillonite exhibiting a stimulatory effect on their degradation (Figure 7). For other phenanthrene series compounds, the natural biodegradation rate by indigenous microorganisms remains low, and the stimulatory effect of montmorillonite is weak (Figure 7).

8. Fluorene series

Although natural biodegradation of fluorene series compounds has indeed been observed, it remains relatively weak. The role of montmorillonite is unclear: both stimulatory and inhibitory effects have been detected (Figure S8). This may be attributed to the low concentration of fluorene series compounds, where the observed fluctuations are reasonable and should not be considered a definitive pattern.

9. Biphenyl series

For biphenyl series compounds, the content trend as: crude oil > M0 ≈ C0 > C753 > M753. Both the natural biodegradation of biphenyl and the stimulatory effect of montmorillonite are clearly demonstrated (Figure 8).

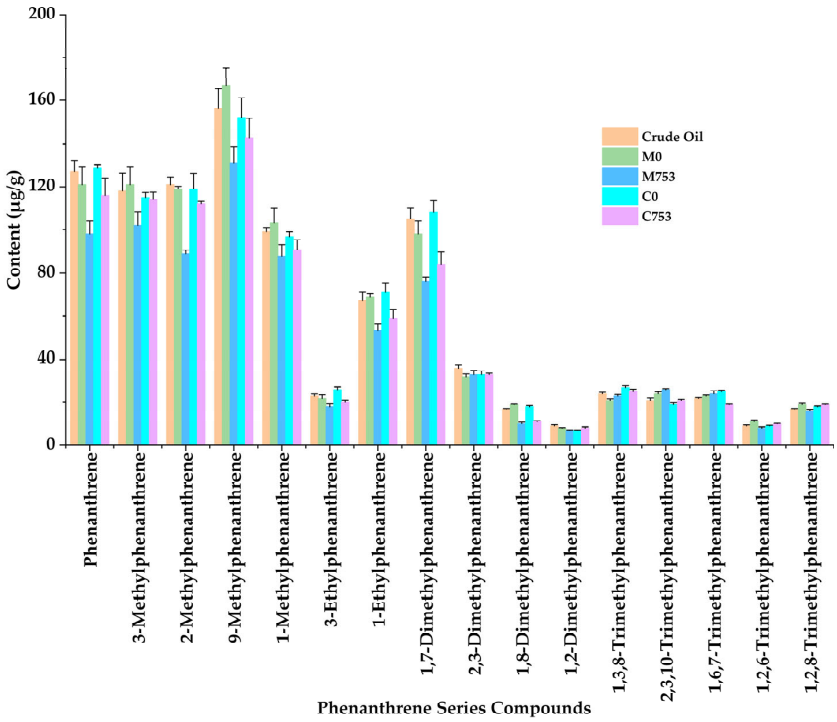

Figure 7. Content of phenanthrene series compounds in crude oil-contaminated soil from well pad before and after field test.

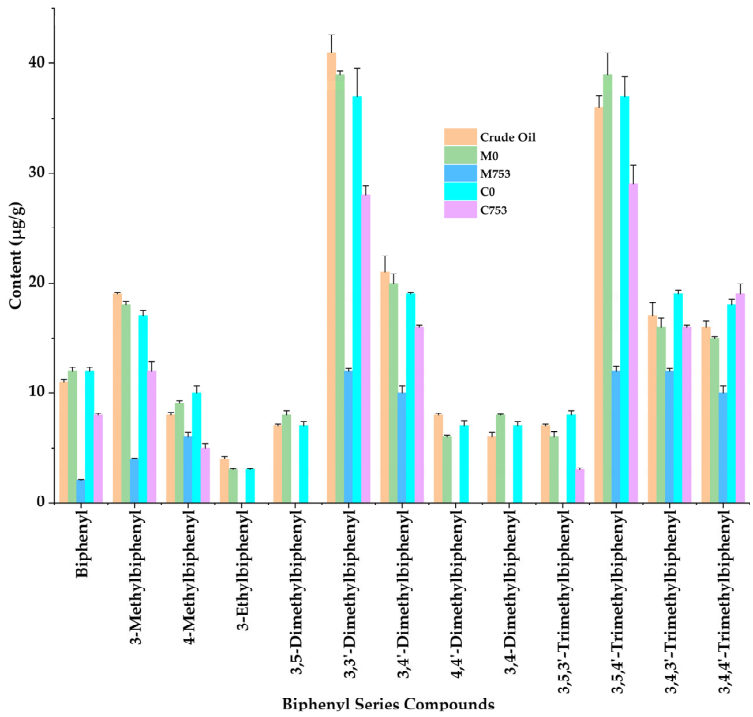

Figure 8. Content of biphenyl series compounds in crude oil-contaminated soil from well pad before and after field test.

10. High-ring number (≥4) aromatic hydrocarbons

Soil indigenous microorganisms can slightly degrade high-ring number aromatic hydrocarbons. The role of montmorillonite in the natural biodegradation of these compounds remains unclear, possibly due to their low concentrations, resulting in no trends (Figure S9).

3.3. Microorganisms in Crude Oil

Microbial analysis of the crude oil identified eight genera: *Stenotrophomonas*, *Desulfotomaculum*, *Pseudomonas*, *Sphingomonas*, *Pseudothermotoga*, *Thermodesulfobacterium*, *Desulfomicrobium*, and *Arcobacter*. Among these, *Arcobacter* exhibited the highest relative abundance (59.4%) (Figure 9).

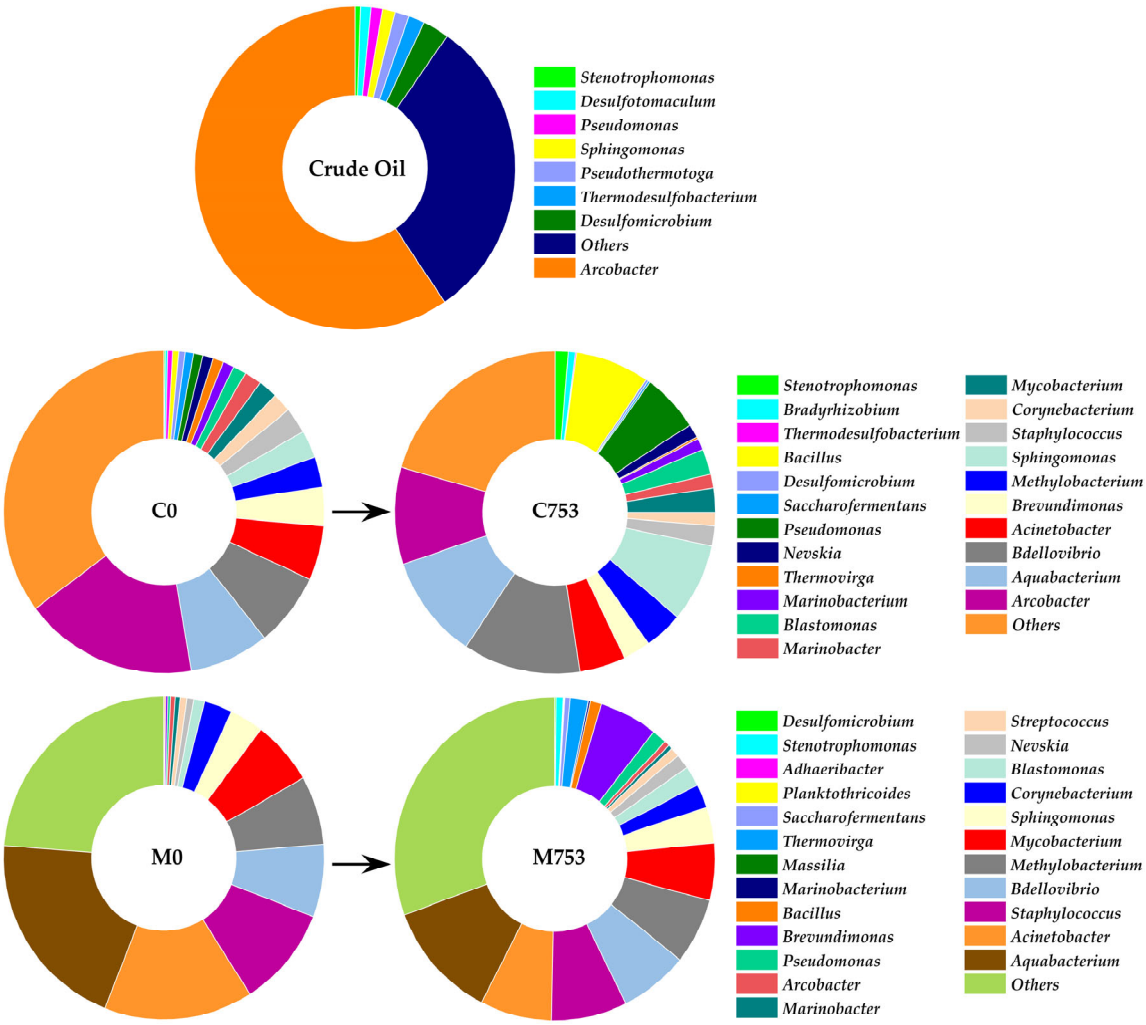

Figure 9. Analysis of microbial gene sequences in crude oil and crude oil-contaminated soil before and after field test.

3.4. Microorganisms in Soil

The linear sampling distance between the test group and control group was approximately 50 m (Figure S1). Although the relative abundance of microbial genera differed between samples M0 and C0, their taxonomic composition remained largely consistent (with 16 genera co-occurring in both samples) (Figure 9). Upon completion of the field test, the DNA concentration in the test group increased from 65 μg/g to 119 μg/g, while that in the control group rose from 77 μg/g to 104 μg/g (Table 1).

3.4.1. Microbial Community in the Control Group

Quantitatively, the microbial genera in the well pad soil without montmorillonite treatment remained unchanged, with 22 identifiable genera (Figure 9). However, over time (753 days), shifts in genus-level abundance were observed. The following genera exhibited marked increases: *Stenotrophomonas* (10.2 folds), *Bradyrhizobium* (3.1 folds), *Bacillus* (12.3 folds), *Pseudomonas* (6.1 folds), and *Sphingomonas* (2.8 folds). Conversely, genera showing notable declines included *Thermodesulfobacterium* (76% reduction), *Desulfomicrobium* (59% reduction), and *Thermovirga* (79% reduction).

Arcobacter consistently dominated the community, yet its abundance decreased by 44% over 753 days (from 17.52% to 9.75%). *Aquabacterium* increased from 8.04% to 10.42%, becoming a new dominant species, while *Bdellovibrio* demonstrated growth (from 7.39% to 11.87%).

3.4.2. Microbial Community in Montmorillonite Treatment Zone

After field test, the number of identifiable microbial genera in the montmorillonite-treated crude oil-contaminated soil increased from 17 to 24 (Figure 9). Newly detected genera included *Desulfomicrobium*, *Stenotrophomonas*, *Adhaeribacter*, *Planktothricoides*, *Saccharofermentans*, *Thermovirga*, and *Massilia*. Notably, the relative abundances of the following genera increased: *Brevundimonas* (26.7 folds), *Pseudomonas* (5.6 folds), and *Bacillus* (8.1 folds). In contrast, *Acinetobacter* and *Staphylococcus* decreased by 52% and 24%, respectively (Figure 9).

4. Discussion

4.1. Biodegradation and Microbial Community in the Control Group

In the control group without montmorillonite treatment, we observed that after 753 days, the soil TOC content decreased by 28.78%, while the crude oil content declined by 33.00% (from 66,981 mg/kg to 44,875 mg/kg) (Table 1). Concurrently, microbial biomass increased, with DNA concentration rising by 35.06% (Table 1). Although volatilization, photolysis, and leaching may have contributed, analysis of climatic conditions (temperature, wind, solar radiation, and precipitation) at the well pad suggests microbial biodegradation was the dominant mechanism.

Over 753 days, the taxonomic composition of soil microbial communities remained unchanged. Neither plowing nor natural evolution under non-anthropogenic intervention significantly altered the microbial species diversity (Table 2). However, a reduction in the abundance of three sulfate-reducing bacteria—*Thermodesulfobacterium* [52], *Desulfomicrobium* [53], and *Thermovirga* [54]—was observed. *Thermodesulfobacterium* and *Thermovirga* are anaerobic thermophiles [55,56], while *Desulfomicrobium* is a mesophilic genus [57]. These taxa are commonly found in the Triassic Yanchang Formation at 65 °C [51] and were likely introduced into the soil via crude oil. Their declining abundance reflects a re-adaptation process to surface soil conditions. Concurrently, aerobic hydrocarbon-degrading genera (*Stenotrophomonas* [58], *Bacillus* [59], *Pseudomonas* [60], and *Sphingomonas* [61]) and the nitrogen-fixing genus *Bradyrhizobium* [62] exhibited marked increases in abundance. The combined decline of thermophilic anaerobes and rise in aerobic hydrocarbon degraders delineates the key characteristics of reservoir-originating microbial communities adapting to surface soil environments.

The microbial community establishes a sophisticated, multi-layered cooperative metabolic network to achieve the complete breakdown of complex petroleum hydrocarbon mixtures. The core principle lies in the complementary integration of aerobic respiration, nitrate reduction, sulfate reduction, and fermentation processes. This enables a comprehensive degradation capacity across both oxic and anoxic environments.

Table 2. Alpha diversity index of samples.

Sample	Shannon-Wiener Index	Simpson Index	Pielou's Evenness Index
Crude Oil	1.642	0.253	0.606
M0	2.485	0.073	0.916
M753	2.476	0.074	0.913
C0	2.318	0.089	0.855
C753	2.501	0.071	0.922

In oxygen-rich zones, aerobic degraders like *Pseudomonas* and *Bacillus* act as pioneers. They employ enzyme systems such as alkane hydroxylase (AlkB) and cytochrome P450 monooxygenases to perform terminal oxidation of straight-chain alkanes and subterminal oxidation of branched-chain alkanes [63]. This process yields corresponding alcohols and aldehydes, which are ultimately channeled into the β-oxidation pathway and the tricarboxylic acid cycle for complete mineralization into carbon dioxide and water. Within this framework, genera like *Sphingomonas* specialize in degrading recalcitrant components such as polycyclic aromatic hydrocarbons. Their unique dioxygenases catalyze the cis-dihydroxylation of aromatic rings, cleaving the benzene ring structure to generate key intermediates like catechol. These intermediates subsequently enter central metabolism via either the ortho- or meta-cleavage pathways. It is noteworthy that *Methylobacterium* may participate in the co-metabolism of aromatic hydrocarbons through homologous enzyme systems of its methane monooxygenase.

As oxygen becomes depleted in localized microenvironments (e.g., inside oil droplets or deep within biofilms), facultative anaerobes such as *Marinobacter* and *Bradyrhizobium* begin to play a dominant role. Under hypoxic conditions, they switch to a nitrate-reducing metabolic mode, utilizing nitrate as an alternative electron acceptor to continue the anaerobic oxidation of alkanes. This achieves a seamless transition at the aerobic-hypoxic metabolic interface. Within strictly anoxic niches, obligate anaerobes like *Desulfomicrobium* and *Thermodesulfobacterium* degrade hydrocarbon classes via the sulfate reduction pathway, a process often accompanied by hydrogen sulfide production. Concurrently, fermentative bacteria like *Saccharofermentans* and *Thermovirga* form the community's internal recycling system. They further ferment long-chain alkanes or the vast quantities of complex intermediates produced during initial degradation. This generates hydrogen, acetate, and short-chain fatty acids. These products serve as crucial substrates and energy sources for other community members, such as sulfate-reducing bacteria. This symbiotic relationship, known as cross-feeding, significantly enhances the overall efficiency of the degradation process.

Crucially, *Bdellovibrio* functions as an "ecological engineer" by preying on dominant populations like *Acinetobacter*. This predatory behavior not only directly controls population sizes, preventing single-species dominance, but also releases sequestered nutrients (e.g., nitrogen, phosphorus) through cell lysis. This promotes nutrient cycling and functional stability within the community. This predator-prey relationship constitutes a cascading regulatory mechanism, serving as a vital safeguard for maintaining long-term degradation activity and functional resilience.

Corresponding to such a microbial community, crude oil microbial degradation primarily targeted saturated hydrocarbons (60.89%) and aromatic hydrocarbons (29.60%), while no changes were observed in resins and asphaltenes. Specifically, n-alkanes and alkylcyclohexanes dominated the saturated fraction, exhibiting clear microbial biodegradation signatures. In contrast, no biodegradation was detected for tricyclic terpanes and hopanes, while only weak natural degradation occurred in bicyclic sesquiterpenes and steranes. The persistence of preferentially degraded hydrocarbons (n-alkanes, alkylcyclohexanes) occurred

alongside minimal utilization of other saturated hydrocarbons, demonstrating sequential degradation characteristics [64]. Within the aromatic fraction, naphthalene and phenanthrene were more abundant than other compounds. Except for trimethylphenanthrene and some dimethylphenanthrenes, biodegradation was evident across all naphthalene and phenanthrene series. Although biodegradation of fluorene, biphenyl, and higher-ring aromatics was detectable, their low initial concentrations resulted in high measurement uncertainties.

4.2. Effects of Montmorillonite on Biodegradation and Microbial Community

In the montmorillonite-treated zone, post-field test measurements revealed greater reductions in soil TOC (33.74%) and crude oil content (37.27%) compared to the control group. Concurrently, DNA concentration increased by 83.08%, substantially exceeding the control group's 35.06% increment. These results demonstrate that montmorillonite exerts a stimulatory effect on both microbial biomass proliferation and crude oil biodegradation.

Regarding microbial taxa, the 753-day field test resulted in increased genus-level diversity within the test zone. Newly detected genera included sulfur-cycling functional taxa (*Desulfomicrobium*), hydrocarbon degraders (*Stenotrophomonas*), biofilm-forming genera (*Adhaeribacter* [65]), and nitrogen-cycling specialists (*Massilia* [66]), along with other genera (*Planktothricoides*, *Saccharofermentans*, and *Thermovirga*). Notably, the abundance of hydrocarbon-degrading genera (*Brevundimonas* [67], *Pseudomonas*, and *Bacillus*) exhibited marked enrichment. The observed increases in microbial biomass and diversity indicate parallel enhancements in functional diversity and metabolic complexity. The proliferation of hydrocarbon-degrading genera directly correlates with intensified crude oil degradation in soil. Compared to the control group, montmorillonite stimulated both microbial biomass accumulation and diversity expansion, which corresponds to the higher crude oil degradation rates recorded in montmorillonite-amended zones.

Under montmorillonite mediation, microorganisms degraded 73.42% of saturated hydrocarbons and 45.77% of aromatic hydrocarbons in soil crude oil, higher than the control group (60.89% and 29.60%, respectively). This definitively demonstrates montmorillonite's stimulatory effect on microbial degradation of both hydrocarbon types. Specifically, montmorillonite stimulated microbial degradation of n-alkanes, alkylcyclohexanes, and dicyclic sesquiterpenes, but showed no stimulation effect on tricyclic terpanes and hopanes. This may be because, in the current system (crude oil and microorganisms), soil microorganisms inherently cannot degrade tricyclic terpanes and hopanes, and montmorillonite cannot stimulate microbial degradation of hydrocarbons that are fundamentally non-degradable in this system. The effect of montmorillonite on sterane biodegradation remains inconclusive. While it stimulated degradation of some steranes, it inhibited others, showing no consistent pattern. This may be because sterane biodegradation itself is relatively weak, and montmorillonite's influence is also minimal, resulting in these irregular observations. For aromatic hydrocarbons, montmorillonite strongly stimulated microbial degradation of naphthalene series compounds, phenanthrene, monomethylphenanthrenes, monoethylphenanthrenes, certain dimethylphenanthrenes (1,7-dimethylphenanthrene), and biphenyl series compounds. For fluorene and higher-ring-number aromatic hydrocarbons, their original concentrations were already low, and biodegradation was inherently weak. Consequently, montmorillonite's effect was not pronounced, leading to complex results where both stimulation and inhibition coexisted without clear patterns.

4.3. Mechanism of Montmorillonite Effects

Montmorillonite exhibits a substantial specific surface area [6], with active sites on its surface capable of adsorbing both organic molecules from crude oil and microorganisms.

This enhanced bioavailability is recognized as one of the key mechanisms underlying montmorillonite's stimulatory effects [14]. Metal cation-enriched montmorillonite can form cation-π bonds, enabling specific adsorption of aromatic hydrocarbons containing aromatic ring structures [13]. Through these mechanisms, montmorillonite collectively promotes the microbial degradation of crude oil.

Simultaneously, we observed that montmorillonite exerted differential effects on structurally analogous organic molecules within the same compound series, as exemplified by steranes, phenanthrenes, fluorenes, and high-ring-number aromatic hydrocarbons. Similar phenomena have also been reported in the illite study [25], likely due to the influence of factors including substituent type and number, alkyl chain length, and aromatic ring count.

Montmorillonite-stimulated crude oil biodegradation likely induces corresponding microbial community shifts along degradation metabolic pathways, potentially explaining the observed increase in microbial diversity. This phenomenon is particularly pronounced in open environmental systems where microbial community enrichment occurs more readily.

4.4. Cost-Effectiveness Analysis

In other well pads within the adjacent area, field tests for remediating petroleum-contaminated soils using various methods were successively conducted. By comparing the cost and effectiveness data of these field tests (Table 3), it was found that the remediation speed of directly adding bacterial agents was faster. Although the montmorillonite stimulation method was the most convenient, it did not have obvious cost advantages. While topsoil replacement yields immediate results, the subsequent disposal cost of contaminated soil is substantial. Combining clay stimulation with highly efficient degrading bacterial strains may emerge as the optimal approach.

Table 3. Cost-effectiveness comparison of field remediation tests for different petroleum-contaminated soils in well pads within the same region.

Method	Well Pad Size (m)	Project and Expenditure (RMB/Yuan)	Total (RMB/Yuan)	Effect
Montmorillonite-mediated Microbial Remediation	48×48	Montmorillonite: 22,870 Plowing equipment: 480 Labor: 600	23,950	TOC decreased by 33.74% over 753 days, crude oil content decreased by 37.27%. Ongoing.
Microbial Remediation	117×104	Bacterial agent: 17,600 Labor: 2400 Maintenance: 18,700	38,700	Crude oil content decreased by 38.9% over 2 years and by 44.6% over 5 years.
Plant-Microbial Synergistic Remediation	100×108	Bacterial agent: 8,900 Plants: 17,690 Labor: 1600 Maintenance: 11,000	39,190	Crude oil content decreased by 21% in the first year. Ongoing.
Topsoil Replacement + Environmental Treatment	132×97	Machinery: 15,700 Labor: 6200 Environmental treatment: 104,600	126,500	Crude oil content decreased by 99% after completion.

5. Conclusions

To the best of our knowledge, this study represents the first report on montmorillonite-mediated microbial degradation of crude oil and microbial community succession in crude oil-contaminated soil from a long-term shutdown (753 days) production well pad. Under natural attenuation (without montmorillonite), crude oil removal was observed, accompanied by increased soil microbial biomass and enriched hydrocarbon-degrading microbial populations. The addition of montmorillonite further stimulated crude oil degradation rates, enhanced microbial biomass, and promoted microbial diversity. Montmorillonite amendment alone achieved cost-effective, operationally simple, and environmentally

friendly bioremediation enhancement. Building upon this foundation, combined strategies (e.g., nutrient supplementation, catalytic additives, or functional microbial consortia) could further improve degradation efficiency and shorten remediation time, demonstrating significant research potential and broad practical applications.

Supplementary Materials: The following supporting information can be downloaded at: https://www.mdpi.com/article/10.3390/environments13010020/s1, Figure S1: Schematic diagram of well pad for field test; Figure S2: SARA composition of crude oil in soil from well pad before and after field test; Figure S3: Field sampling photo of the authors and Yanchang Oilfield staff at the well pad; Figure S4: Total ion chromatogram of saturated hydrocarbons in crude oil-contaminated soil from well pad before and after field test; Figure S5: Content of hopanes in crude oil-contaminated soil from well pad before and after field test; Figure S6: Content of steranes in crude oil-contaminated soil from well pad before and after field test; Figure S7: Total ion chromatogram of aromatic hydrocarbons in crude oil-contaminated soil from well pad before and after field test; Figure S8: Content of fluorene series compounds in crude oil-contaminated soil from well pad before and after field test; Figure S9: Content of high-ring number ($\geq$4) aromatic hydrocarbons in crude oil-contaminated soil from well pad before and after field test.

Author Contributions: Methodology, formal analysis, investigation, data curation, writing—original draft, writing—review and editing, L.L.; Conceptualization, resources, writing—review and editing, project administration, funding acquisition, C.Z.; Writing—review and editing, resources, Y.Z. All authors have read and agreed to the published version of the manuscript.

Funding: This research was funded by the National Natural Science Foundation of China (No. 52170096), the Erdos City Science and Technology Cooperation Major Project (No. 2022EEDSKJZDZX015–2), and the Fundamental Research Funds for the Central Universities (Top Innovative Talents Fund of CUMTB) (No. BBJ2024051).

Data Availability Statement: Data will be made available on request.

Acknowledgments: The primary content of this study is derived from the doctoral dissertation of the corresponding author, Li Lei, completed at the China University of Petroleum (Beijing), under the supervision of Wan Yunyang. We acknowledge the staff of Shaanxi Yangchang Petroleum (Group) Co., Ltd. for providing the data and all assistance during the sampling process. We also thank Li Weiheng and Tian Xiukun for their personal financial assistance in the initial stage.

Conflicts of Interest: Author Yue Zhang was employed by the CNOOC Gas & Power Group, Research & Development Center. The remaining authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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